Figure 2.11 The important criteria to consider when consuming an LLM

For our purposes of building AI agents, we need to look at each of these criteria in terms related to the task. Model context size and speed could be considered the sixth and seventh criteria, but they are usually considered variations of a model deployment architecture and infrastructure. An eighth criterion to consider for an LLM is cost, but this depends on many other factors. Here is a summary of how these criteria relate to building AI agents:

- *Model performance*—You'll generally want to understand the LLM's performance for a given set of tasks. For example, if you're building an agent specific to coding, then an LLM that performs well on code will be essential.
- *Model parameters (size)*—The size of a model is often an excellent indication of inference performance and how well the model responds. However, the size of a model will also dictate your hardware requirements. If you plan to use your own locally hosted model, the model size will also primarily dictate the computer and GPU you need. Fortunately, we're seeing small, very capable open source models being released regularly.
- *Use case (model type)*—The type of model has several variations. Chat completions models such as ChatGPT are effective for iterating and reasoning through a problem, whereas models such as completion, question/answer, and instruct are more related to specific tasks. A chat completions model is essential for agent applications, especially those that iterate.
- *Training input*—Understanding the content used to train a model will often dictate the domain of a model. While general models can be effective across tasks, more specific or fine-tuned models can be more relevant to a domain. This may be a consideration for a domain-specific agent where a smaller, more fine-tuned model may perform as well as or better than a larger model such as GPT-4.
- *Training method*—It's perhaps less of a concern, but it can be helpful to understand what method was used to train a model. How a model is trained can affect its ability to generalize, reason, and plan. This can be